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MARKETING | REVIEW ARTICLE

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Data security and privacy concerns of AI-driven marketing in the context of economics and business field: an exploration into possible solutions

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ABSTRACT
Interest in artificial intelligence (AI) is widespread across several industries, such as marketing, but worries about its ethical and legal consequences are increasing. This article examines concerns regarding data security and privacy in AI-powered marketing and discusses possible remedies. The study compiles information from academic articles using a comprehensive literature review. The key conclusions emphasise issues including data confidentiality, distribution, cyberattacks, fraud, and disinformation. Addressing these concerns involves providing privacy insurance, improving technology readiness, enforcing security regulations, and building regulatory frameworks. The report emphasises the need for transparency in the use of AI by marketing professionals, highlighting the need to keep clients aware of data practices. This research establishes the foundation for future investigation, encouraging continuous discussion and examination of this developing topic.

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SUBJECTS
Artificial Intelligence; Business, Management and Accounting; Economics

1. Introduction

Artificial intelligence (AI) is an expansive domain of computer science focused on creating systems and algorithms capable of executing activities that usually necessitate human intelligence (Duan et al., 2019; Feng et al., 2021). These tasks involve a diverse array of activities, such as problem-solving, speech recognition, learning, and decision-making (Duan et al., 2019; Song & Ma, 2010). The objective of AI is to develop computers capable of emulating cognitive activities, allowing them to adapt and enhance their performance progressively (Huang & Rust, 2021). The origins of AI can be traced back to ancient civilizations, where humanoid robots were depicted in myths and stories. Nonetheless, the establishment of AI as an academic field commenced throughout the middle of the 20th century (Tobin et al., 2020). The phrase 'artificial intelligence' was first introduced in 1956 at the Dartmouth Conference, which marked the initiation of focused research in this domain (Franklin, 2014; Tobin et al., 2020). Initially, AI implementations mostly concentrated on rule-based systems and symbolic reasoning (Duan et al., 2019; Lu, 2019). However, the area has experienced a significant transformation with the emergence of machine learning and deep learning technologies. Currently, AI plays a crucial role in numerous industries, fundamentally transforming the automation of work, the resolution of problems, and the ability of systems to adjust to changing settings (Chintalapati & Pandey, 2022).

The introduction of advanced AI technologies has greatly revolutionised the marketing industry specifically in the context of economics and business, improving decision-making, personalised targeting, and overall campaign efficiency. AI in marketing is the incorporation of artificial intelligence methods

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and tools to enhance many aspects of the marketing process (Chintalapati & Pandey, 2022; Verma et al., 2021). This spans a wide range of applications, including machine learning techniques, natural language processing, and data analytics. The main goal is to utilise these technologies to examine extensive datasets, comprehend customer behaviour, and extract practical insights that can inform and enhance marketing efforts. The function of AI in marketing encompasses customer segmentation, predictive analytics, real-time engagement through chatbots, recommendation systems, and the automation of repetitive processes. Through the use of artificial intelligence, marketers can generate more precise and customised experiences for their target audiences, ultimately leading to enhanced engagement and conversion rates (Kumar et al., 2019).

The integration of AI technology in marketing presents substantial difficulties and potential within the current economic and corporate landscape (Dwivedi et al., 2021; Loureiro et al., 2021). AI facilitates the utilisation of extensive data to achieve more precise market segmentation, personalised messaging, and optimised marketing campaigns, hence enhancing the efficiency and efficacy of marketing initiatives (Haleem et al., 2022). Nevertheless, the issue of data security and privacy in AI-driven marketing poses a complex challenge in the fields of economics and business (Kunz & Wirtz, 2024; Schlögl et al., 2019). AI technologies empower marketers to utilise extensive collections of consumer data for precise advertising and customised experiences. AI has changed consumer behavior allowing them to shop from home without having to visit a supermarket. This automation influences the recruitment of new roles for Human Resources (HRs) to build competitive advantage through the utilization of AI to improve the efficiency of companies in responding to consumer needs (Li et al., 2023).

However, the implementation of AI also give rise to substantial concerns over the safeguarding of sensitive information (Schlögl et al., 2019). Data breaches, unauthorised access, and misuse have a significant potential to undermine consumer trust and pose a risk of reputational damage for businesses (La Torre et al., 2018). Furthermore, with the advancement of AI algorithms in analysing consumer behaviour, there is an escalating concern about violating individual privacy rights, which could lead to increased regulatory scrutiny and legal consequences (Bandara et al., 2020). These worries not only inhibit the implementation of AI-driven marketing tactics but also have broader economic repercussions, as they might discourage customer participation, obstruct innovation, and result in significant financial losses.

While research on AI marketing's effectiveness is abundant, a gap exists regarding the specific security and privacy implications within the economic and business context. Existing literature often focuses on technical aspects of data security or explores privacy concerns from a purely legal standpoint. This systematic review aims to bridge this gap by examining how these issues impact businesses economically and strategically. It will delve into the potential financial consequences of data breaches for businesses using AI marketing, analyzing lost customer trust, brand damage, and potential regulatory fines. Additionally, the review will explore the impact on business strategy, investigating how privacy concerns might limit data collection and personalization, potentially hindering the effectiveness of AI marketing campaigns.

Current research offers valuable insights into potential solutions, but a comprehensive framework for navigating this complex landscape is lacking. This review will analyze existing solutions proposed in academic literature, regarding the use of AI in marketing practices while protecting consumer privacy by examining current research on data security and privacy tactics. This method allows stakeholders to utilise AI in marketing efficiently while also adhering to privacy standards, ultimately improving economic advantages and client confidence. This research evaluation provides guidance to decision-makers in creating ethical and enduring marketing tactics.

2. Methodology

2.1. Research design

In this research study, we aim to examine current gap in understanding the precise security and privacy consequences of AI marketing in economic and business contexts. To achieve this goal, a systematic literature review was conducted to comprehensively analyze existing literature and propose insights into

the intersection of AI marketing, security, and privacy. The primary research objectives of this systematic literature review are to answer these research questions:

- What are the primary challenges and implications for businesses utilizing AI marketing strategies, and how do these concerns impact consumer trust and regulatory compliance?
- What strategies and frameworks can effectively address data security and privacy concerns in AI-driven marketing practices, and how do these solutions impact business operations, consumer trust, and regulatory compliance?

2.2. Data source and data collection

This study employed a systematic literature review (SLR) strategy to thoroughly investigate existing research on the selected issue. This strategy entails systematically collecting, merging, and evaluating relevant material to provide valuable insights that direct the research process and improve understanding of the issue. Researchers followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) approach as described by Moher et al. (2009). The PRISMA technique comprises four stages: identification, screening, eligibility, and inclusion (Moher et al., 2009; ter Huurne et al., 2017). Figure 1 shows the PRISMA flow chart outlining the several stages of this study.

To enhance the comprehensiveness and replicability of this SLR, a detailed methodology was rigorously employed. The PRISMA flow diagram served as a guiding framework, ensuring transparency and consistency throughout the review process. The initial step involved the identification of relevant databases and papers to establish a robust foundation for the review. A systematic search strategy was devised utilizing five strategically chosen keywords: 'data security', 'privacy concern', 'marketing', and the synonymous terms 'Artificial intelligence' and 'AI'. This search string encompassed key concepts directly relevant to the research objectives. The search was conducted within the Scopus databases to ensure the inclusion of high-quality peer-reviewed research.

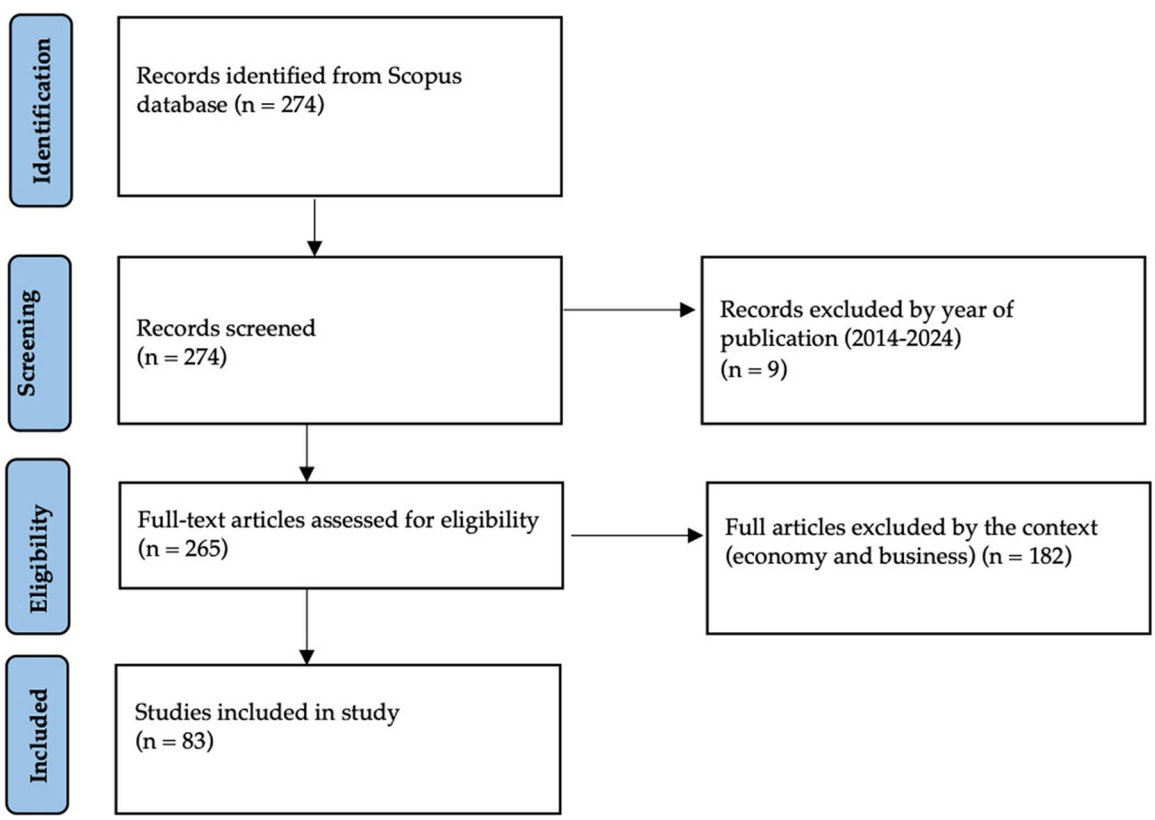

Figure 1. Prisma flow diagram for SLR.

Several inclusion and exclusion criteria were implemented to select papers that aligned with the predefined research parameters. Firstly, English language publications were prioritized to ensure accessibility and facilitate understanding among the intended audience. Additionally, the publication timeframe was restricted to January 2014 to February 2024, reflecting the contemporary relevance of the topic and enabling the examination of recent advancements and trends in the field. Furthermore, papers were required to be thematically relevant to the domains of economy and business, mirroring the specific focus of the review on AI-driven marketing and its implications for organizational strategies and market dynamics. These criteria were meticulously chosen to ensure the relevance and coherence of the selected literature with the overarching research objectives.

3. Results and discussion

3.1. Prevailing data security and privacy concerns

3.1.1. Privacy concerns: cyberattack, scam, and falsification of information

The importance of dealing with privacy issues in AI-powered marketing within the economic and corporate landscape is crucial, considering the rapid increase in data gathering and use for personalized advertising and customer profiling (Breward et al., 2017; Cheng et al., 2023; Coss & Dhillon, 2019; Hille et al., 2015; Tiwari et al., 2023; Wiegard & Breitner, 2019; Wieringa et al., 2021; Wu et al., 2024). The growing dependence on AI algorithms to analyze large quantities of personal data has made privacy breaches and information misuse a crucial concern (Franz & Benlian, 2022). Cyberattacks pose a major privacy threat by exploiting weaknesses in AI systems to obtain sensitive customer data without authorization (Pérez-Morón, 2022). Sophisticated phishing assaults on marketing databases can result in the exposure of personal information, such as financial details and surfing history, which poses significant risks to consumer privacy and confidence (Nikkhah et al., 2024).

AI-driven marketing offers a privacy danger through schemes involving health record devices and human-tracking technologies, leading to the exploitation of personal health data for illicit activities (Dogra et al., 2023). Misleading marketing tactics may deceive individuals by distorting health information from wearable devices or medical records, leading to financial abuse or identity theft (De Moya et al., 2021; Matt et al., 2019). These breaches infringe on privacy rights and diminish trust in AI-powered marketing efforts, weakening customer confidence in online commerce systems.

The falsification of information is a significant privacy issue in AI-driven marketing. Deceptive methods, including data tampering (Acquisti et al., 2016; Cloarec, 2022) and misinformation campaigns, are used to influence consumer behavior and purchasing decisions (Choi et al., 2020). False advertising claims and fake customer testimonials can deceive consumers, leading to a loss of trust in the genuineness of promoted items or services. This can cause reputational harm to organizations and financial setbacks. The combined effect of these privacy challenges on economic growth is significant, since they reduce customer trust, discourage investment in AI technology, and require governmental actions to protect data privacy and reduce risks. The lack of proper safeguards and ethical norms in AI-driven marketing leads to several privacy problems, hindering innovation, market efficiency, and sustainable economic growth.

3.1.2. Data security: data confidentiality and data sharing

Data security is a crucial issue in economic and corporate operations, particularly in AI-powered marketing (Maiorescu et al., 2021; Wu et al., 2024). Data security is the practice of safeguarding digital information from unauthorized access, alteration, or theft at every stage of its existence. The AI-driven marketing environment significantly depends on analyzing large volumes of customer data for targeted advertising and personalized marketing techniques, which presents a major difficulty in maintaining data security (Rosário & Dias, 2023). Businesses need to protect sensitive consumer data while using AI algorithms for analysis and revenue generation. This requires strong encryption, access controls, and threat detection to reduce cyber risks and breaches.

The healthcare industry faces a significant data security threat, especially concerning data confidentiality (Ermakova et al., 2020). Malevolent individuals might target the extremely sensitive information held in

patient records, making the confidentiality of medical data crucial. Violations of medical data confidentiality not only infringe upon patient privacy rights but also have significant ethical and legal consequences (Wadmann et al., 2023). Unauthorized access to medical data can result in identity theft, insurance fraud, or compromise patient safety (Javaid et al., 2023). This highlights the continuous struggle for healthcare organizations to uphold the integrity and confidentiality of medical data by implementing strict security measures and complying with regulatory frameworks.

Moreover, the sharing of client data between firms adds another level of data security risk. Data sharing strategies facilitate collaboration and creativity but can give rise to worries about data privacy and protection (Chatterjee et al., 2023; Hayes et al., 2020; Saura et al., 2021; Xiao et al., 2023). Companies must manage intricacies to ensure permission, openness, and accountability when dealing with customer data. Unauthorized access or exploitation of shared data result in breaches of trust, reputational harm, and regulatory fines, highlighting the importance of strong data security processes and risk management strategies (Habbal et al., 2024).

3.2. Exploration for possible solutions

Pérez-Morón (2022) identified five elements that influence the improvement of data security and customers' privacy: relative advantage, technological readiness, top management support, firm size, and government policy and regulations. Based on a comprehensive literature review, several solutions have been proposed to address data security and privacy challenges:

3.2.1. Relative advantage through customers' privacy insurance

One effective way to deal with data security and privacy sharing issues is by utilizing the idea of relative advantage with client data insurance (Cheng et al., 2023; Wiegard & Breitner, 2019). Businesses can enhance client safety by providing customer data insurance, reducing financial and reputational risks linked to data breaches or unauthorized access. This method encourages firms to invest in strong data security measures and also builds trust with customers by showing a dedication to protecting their sensitive information. Customer data insurance benefits organizations and customers by encouraging investments in data security and providing financial protection in case of security problems (Cheng et al., 2023). Organizations can use this solution to address data security concerns, build trust and loyalty, and strengthen their competitive advantage in the marketplace. In implementation, insurers should collaborate with device manufacturers who can be assigned responsibility for ensuring data security and transmission and increasing transparency regarding data usage, preventing theft and manipulation of data by third parties (Wiegard & Breitner, 2019).

3.2.2. Technological readiness through RFID and HITAM

One effective approach to improving data security and privacy sharing is by utilizing radio frequency identification (RFID) (Tang et al., 2019) and the information technology acceptance model (HITAM) (De Moya et al., 2021; Matt et al., 2019), especially in the realm of health monitoring gadgets. RFID technology facilitates the effective monitoring and control of confidential medical information, guaranteeing safe transfer and retention during the device's lifespan. Manufacturers can enhance security in health tracking devices by including RFID tags and implementing encryption and authentication processes to safeguard patient data from unauthorized access or modification. For example, Tang et al. (2019) used RFID to positively contribute to the perceived security and privacy felt by professionals. This is because RFID is able to help minimize human error, thereby increasing their organizational efficiency.

Additionally, De Moya et al. (2021) and Matt et al. (2019) reported that HITAM provides information on user acceptability and adoption of security-enhancing technologies, aiding in the smooth integration of data security measures into health tracking devices. HITAM allows developers to pinpoint customer preferences, concerns, and usability needs to ensure that security features are user-friendly and meet consumer expectations. Organizations can improve data security and privacy on health tracking devices by using RFID technology and HITAM. This can help build trust among users and encourage the widespread adoption of these advanced technologies in healthcare settings.

3.2.3. Top management support and company size: implementing multilateral security requirements

Securing top management support by implementing multilateral security criteria based on firm size are crucial steps to overcome data security and privacy sharing concerns (Ermakova et al., 2020). Integrate multilateral security requirements analysis into the requirements engineering process to fully consider interacting security demands across different organizational functions. This strategy entails involving senior management in supporting and ranking data security projects while customizing security protocols to suit the organization's scale and operational intricacies. To implement multilateral security requirements, a comprehensive evaluation of security needs in many departments is necessary (Ermakova et al., 2020). This involves identifying possible vulnerabilities and creating a unified framework to deal with them. The framework should include strong encryption techniques, access controls, personnel training programs, and routine security audits to guarantee compliance and efficiency. Businesses can proactively reduce data security risks, promote security awareness, and protect sensitive information by obtaining high-level support and tailoring security procedures to match the organization's size (Khalifa Alhitmi et al., 2023).

3.2.4. Policy and regulation: mitigation, policy and framework development

An effective way to address concerns about data security and privacy sharing is through the implementation of policies and regulations that attempt to minimise perceived risks and improve protection. Policies for cashless payment systems can establish strict security rules and processes to protect financial transactions and prevent unauthorised access to sensitive payment data (Rahman et al., 2020). Customised privacy policies and procedures for cloud computing services can guarantee the safe storage and transfer of data by establishing explicit rules for data management and access restrictions to safeguard user privacy (Coss & Dhillon, 2019). Furthermore, creating privacy-preserving service frameworks can provide universal standards for data anonymization and encryption in the industry, enabling secure data sharing among organisations while reducing the chances of unauthorised exposure (Piao et al., 2016).

Implementing ethics frameworks in industries such as tourism and hospitality can encourage responsible data management by prioritising openness, accountability, and user consent in the gathering and utilisation of personal data (Yallop et al., 2023). Businesses and regulatory agencies may work together to establish strong policies and laws that provide a secure and ethical environment for data sharing. This collaboration will enhance customer trust and confidence in digital transactions and services. For example, Wieringa et al. (2021) identified five responsibilities that must be considered when analyzing consumer personal data: data collection, data verification, data storage and control, insight generation and insight dissemination. These five responsibilities can be implemented at three levels (customers, intermediaries and companies).

Utilising policy and law to address data security and privacy sharing concerns requires a thorough approach that covers many areas of risk reduction and adherence to rules. Risk mitigation tactics might include performing comprehensive risk assessments to detect vulnerabilities and applying solutions like encryption, data masking, and access controls to reduce these risks (Chawla et al., 2023). Enforcing privacy rules and practices for cloud computing services guarantees that data stored and processed in the cloud stays secure and is compatible with relevant privacy regulations, such as the General Data Protection Regulation (GDPR). Moreover, the advancement of privacy-preserving service frameworks promotes the use of privacy-enhancing technology and practices, cultivating a culture of data protection and privacy awareness among both enterprises and customers (Maiorescu et al., 2021). Implementing ethics frameworks in the tourism and hospitality sectors encourages ethical data practices and responsible use of customer data, which improves consumer trust and loyalty. By incorporating these policy and regulatory measures, organisations can reduce data security risks, safeguard consumer privacy, and establish confidence and integrity in data sharing procedures.

4. Conclusions

Addressing data security and privacy concerns in AI-driven marketing necessitates a comprehensive strategy involving technological advancements, organisational tactics, and regulatory guidelines. Businesses

can encourage investments in strong data security measures and improve consumer trust and loyalty by using customer data insurance to capitalise on relative advantages. Implementing Radio Frequency Identification (RFID) and the Health Information Technology Acceptance Model (HITAM) in health tracking devices demonstrates technological maturity and provides practical ways to enhance data security and privacy sharing. Securing senior management support and customising global security requirements according to firm size are crucial elements in effectively handling data security concerns. Policy and legislation are crucial in reducing perceived risks and improving data protection by implementing privacy-preserving service frameworks and ethics frameworks across different industries. Organisations may use these technologies to effectively manage data security and privacy sharing, which will help build trust, encourage innovation, and promote sustainable economic growth in the digital age. While this SLR provides valuable insights into addressing data security and privacy concerns in AI-driven marketing, it underscores the need for further research to delve deeper into several key areas. Future investigations could focus on the development and validation of comprehensive frameworks integrating technological advancements, organizational tactics, and regulatory guidelines to bolster data security measures and enhance consumer trust.

Authors' contributions

Conception and design (Hitmi Khalifa Alhitmi and Alin Mardiah); analysis and interpretation of the data (Alin Mardiah); the drafting of the paper, revising it critically for intellectual content (Alin Mardiah, Khalid Ibrahim Al-Sulaiti, and Jaffar Abbas); and the final approval of the version to be published (Hitmi Khalifa Alhitmi). All authors agree to be accountable for all aspects of the work.

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Data availability statement

All data (papers) analysed are included in Scopus database.

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